Lecture 10.

Thursday, 21 September 2023 10:37 AM

Temp. programmed Surface reactions

TPR → Temp. prog. reduction

Reduction mechanisms

- Shrinking core (H_2 fast dissociation)
- Nucleation & growth (H_2 dissociation slow) (stable oxides)

} formed an oxide & now need to reduce it from metal (+ water)

+ Linear ramp rate
+ H_2 consumption

} Experimentally?

$dT = \beta dt$

$r \propto \beta$

[TCD or MS] → to measure H_2 consumption rate

1) Assumption → model does not correspond w/ temp ramp → Is that true → does a particular oxide reduce in a particular way.

You can resolve peaks by using wet & dry reduction to shift equilibriums as recorded.

TGA - DSC

↑ tells you if the weight loss was exothermic or endothermic

TPSR →

Hydrocarbons serve as reductants for NO_x

i.e. provide H to break / reduce NO_x